7 **(a)** **(i)** For a fixed tension (or load) in the string, the speed of the transverse progressive wave on the string is constant.

For stationary wave to be seen, the length of the string must be integer multiples of half-wavelength of the wave.

Since the length of the string is fixed, this means that only certain wavelengths and hence frequencies will produce observable stationary waves.

(ii) Since length of string L must be integer multiples of half-wavelength,

$$L = n \times \tfrac{1}{2}\lambda_n = n \times \tfrac{1}{2} \cdot \frac{v}{f_n} \qquad (v = f\lambda)$$

where n, the number of segments, is an integer.
Rearranging,

$$f_n = n\frac{v}{2L} \text{ (shown)}$$

(iii) $n = 4$ for a stationary wave with 5 nodes

$$v = \frac{2Lf_n}{n} = \frac{2 \times 0.600 \times 40.0}{4} = 12.0 \text{ m s}^{-1}$$

(iv) $f_n \propto v \propto T^{1/2}$

Solution 1

$$\frac{f_n'}{f_n} = \sqrt{\frac{T'}{T}} = \sqrt{0.98}$$

$$f_n' = \sqrt{0.98} f_n = \sqrt{0.98} \times 40.0 = 39.598 = 39.6 \text{ Hz}$$

Solution 2
Since $f_n \propto v \propto T^{1/2}$,

$$\frac{\Delta f_n}{f_n} = \frac{1}{2}\frac{\Delta T}{T} = \frac{1}{2} \times (-0.02) = -0.01$$

$$\Delta f_n = -0.01 f_n = -0.01 \times 40.0 = -0.4 \text{ Hz}$$

New frequency

$$f_n' = f_n + \Delta f_n = 40.0 + (-0.4) = 39.6 \text{ Hz}$$

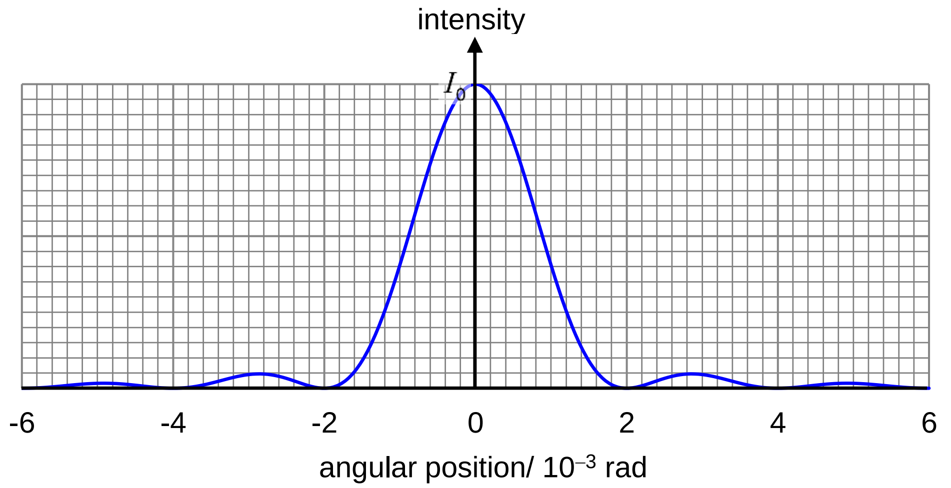
(b) **(i)** **1.** Using $b\sin\theta_m = m\lambda$ where $m = 1$

$$\theta_1 = \sin^{-1}\left(\frac{\lambda}{b}\right)$$

$$= \sin^{-1}\left(\frac{600 \times 10^{-9}}{0.30 \times 10^{-3}}\right)$$

$$= 2.0 \times 10^{-3} \text{ rad (shown)}$$

2.



3. For the two point sources to be just resolved,

$$\theta_1 = \theta_{\min} = 2.0 \times 10^{-3}$$

$$\frac{0.50}{D} = 2.0 \times 10^{-3}$$

$$D = \frac{0.50}{2.0 \times 10^{-3}} = 250 \text{ m}$$

The distance of the sources from the slit is 250 m.

(ii) **1.** Separation between 2 slits

$$a = 0.30 - 0.10 = 0.20 \text{ mm}$$

2. first minimum of single slit diffraction envelop:

$$\theta_1 = \sin^{-1}\left(\frac{\lambda}{b}\right) = \sin^{-1}\left(\frac{600 \times 10^{-9}}{0.10 \times 10^{-3}}\right) = 6.0 \times 10^{-3} \text{ rad}$$

single slit minima: $b \sin\theta_m = m\lambda$ --- (1)
double slit maxima: $a \sin\theta_n = n\lambda$ --- (2)

for $\theta_m = \theta_n$, $\frac{(1)}{(2)}$: $\frac{b}{a} = \frac{m}{n}$

missing orders of maxima of double slit interference pattern

$$n = m\frac{a}{b} = m\frac{0.20}{0.10} = 2m$$

An interference pattern is observed with <u>maxima occurring at intervals</u> of <u>3.0×10^{-3} rad</u>.

The pattern is modulated by a diffraction envelop with <u>minima occurring at intervals</u> of <u>6.0×10^{-3} rad</u>.

<u>Even orders of interference maxima are missing</u>.

3. Before applying the film, assume that the amplitude of the light at the central maximum is A_0, then $I_0 = kA_0^2$.

Amplitude at the central maximum due to each slit is now $A_0/3$.
(1/3 the number of original waves arrive at the central maximum)

Amplitude at central maximum due to both slits is $2A_0/3$.
(superposition of waves from each slit)

intensity of central maximum

$$I = k\left(\frac{2A_0}{3}\right)^2 = \frac{4}{9}\cdot kA_0^2 = \frac{4}{9}\cdot I_0$$

8 **(a)** The number 238 refers to the number of nucleons in the nucleus or number of protons